

Machine Intelligence

4. Reasoning Under Uncertainty, Part II: Inference in Bayesian networks

How to Answer your (Probabilistic) Questions

Álvaro Torralba



AALBORG UNIVERSITET

Fall 2024

Reminder: Recovering the Full Joint Probability Distribution

“A Bayesian network is **a methodology for representing the full joint probability distribution.**”

→ How to recover the full joint probability distribution $\mathbf{P}(X_1, \dots, X_n)$ from $BN = (\{X_1, \dots, X_n\}, E)$?

Chain rule: For **any** ordering $X_1, \dots, X_n$, we have:

$$\mathbf{P}(\mathbf{X}_1, \dots, \mathbf{X}_n) = \mathbf{P}(\mathbf{X}_n \mid \mathbf{X}_{n-1}, \dots, \mathbf{X}_1) \mathbf{P}(\mathbf{X}_{n-1} \mid \mathbf{X}_{n-2}, \dots, \mathbf{X}_1) \dots \mathbf{P}(\mathbf{X}_1)$$

Choose $X_1, \dots, X_n$ **consistent with BN**: $X_j \in \text{Parents}(X_i) \implies j < i$.

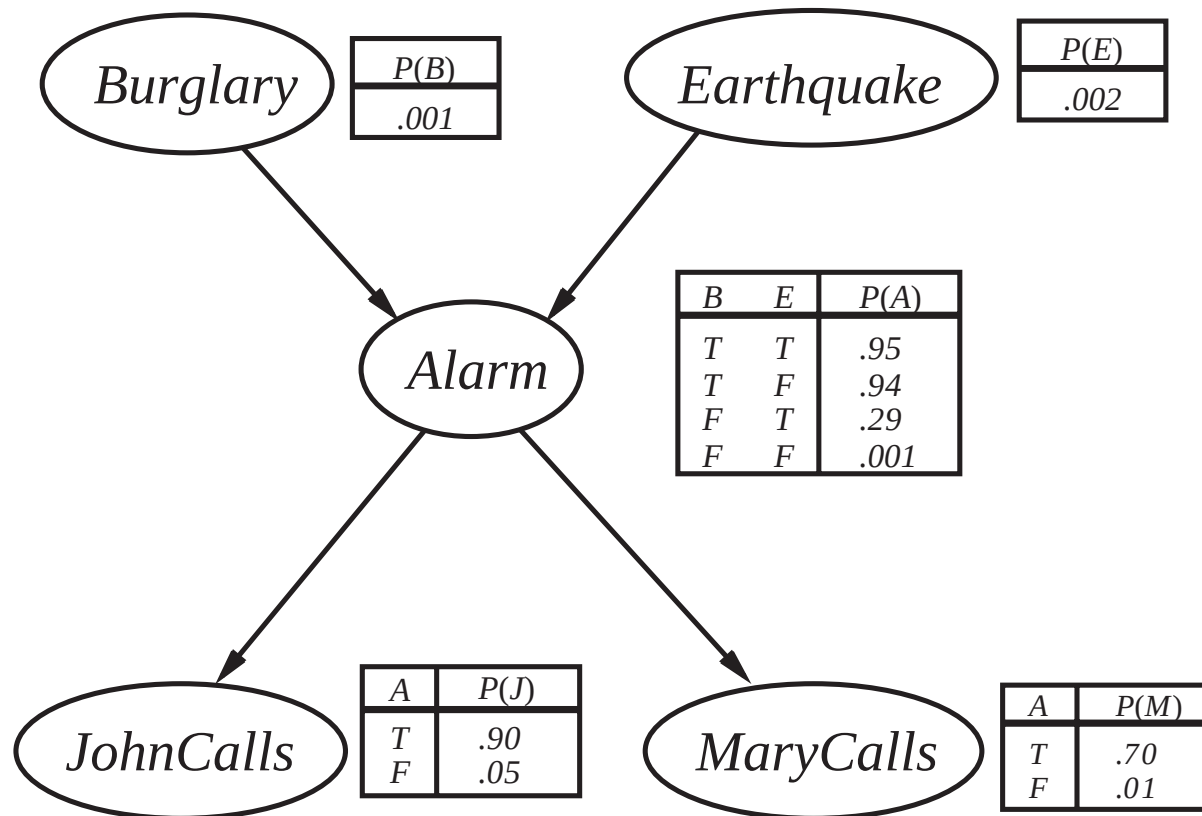
Exploit conditional independence: With **BN assumption (A)**, instead of $\mathbf{P}(X_i \mid X_{i-1} \dots, X_1)$ we can use $\mathbf{P}(X_i \mid \text{Parents}(X_i))$:

$$\mathbf{P}(\mathbf{X}_1, \dots, \mathbf{X}_n) = \prod_{i=1}^n \mathbf{P}(\mathbf{X}_i \mid \text{Parents}(\mathbf{X}_i))$$

The distributions $\mathbf{P}(X_i \mid \text{Parents}(X_i))$ are given by **BN assumption (B)**.

→ Same for atomic events $P(x_1, \dots, x_n)$.

Reminder: Recovering a Probability for John, Mary, and the Alarm



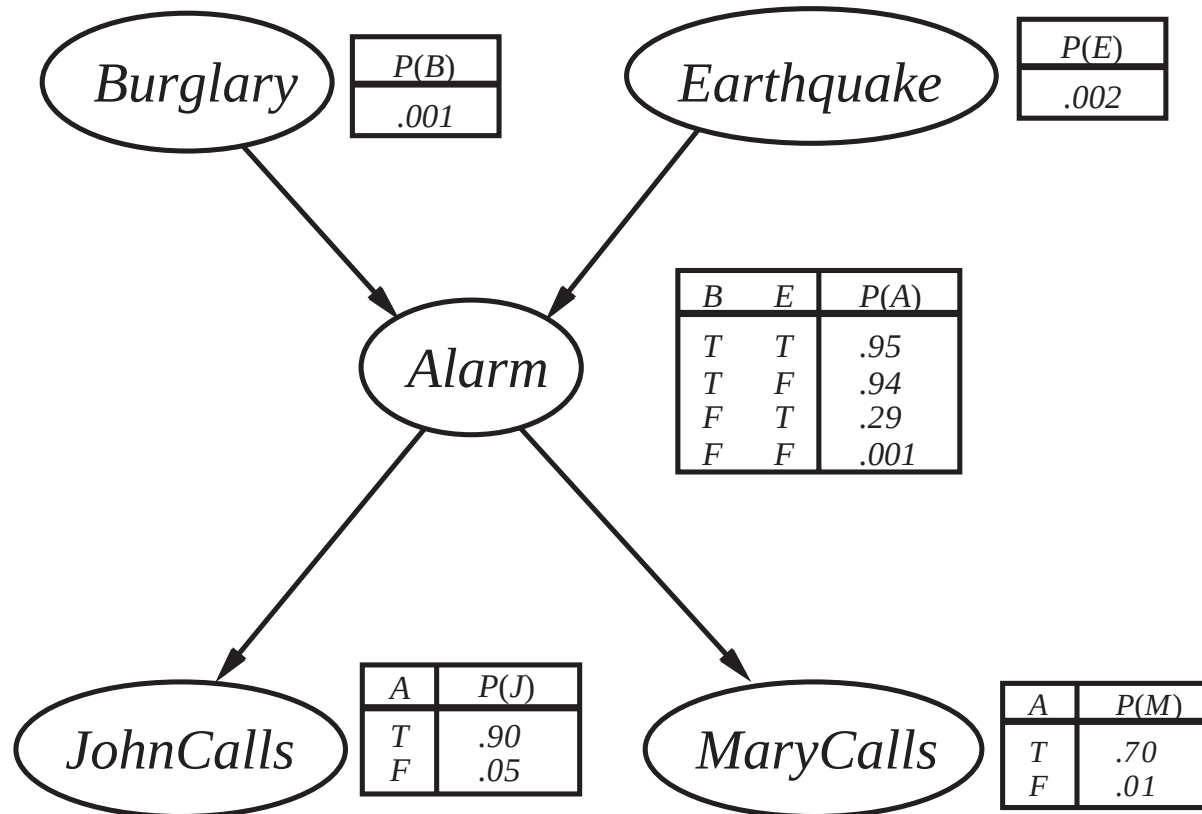
$$\begin{aligned}
 \mathbf{P(j, m, a, \neg b, \neg e)} &= P(j \mid a) \cdot P(m \mid a) \cdot P(a \mid \neg b, \neg e) \cdot P(\neg b) \cdot P(\neg e) \\
 &= 0.9 * 0.7 * 0.001 * 0.999 * 0.998 \\
 &= 0.000628
 \end{aligned}$$

Our Agenda for This Chapter

- **Probabilistic Inference Tasks:** What problems do we want to solve?
→ **Before doing algorithms that answer, we need to understand what the questions are about!**
- **Exact Inference: Naive Enumeration:** A basic method to solve the probabilistic inference tasks
→ **How to obtain the exact answer to our questions !**
- **D-Separation:** How evidence is transmitted along the Bayesian Network?
→ **Some intuition about how BNs work.**
- **Approximate Inference via Sampling:** What to do when computing the exact answer is too expensive?
→ **We can approximate the solution via sampling methods!**
- **Further Material: Exact Inference: Variable Elimination:** A more clever way of using the structure of the Bayesian Network to our advantage! → Not exam relevant

Inference for Mary and John

→ Observe **evidence variables** and draw conclusions on **query variables**.



What is $P(\text{Johncalls})$?

What is $P(\text{Burglary} \mid \text{johncalls})$?

What is $P(\text{Burglary} \mid \text{johncalls}, \text{marycalls})$?

Probabilistic Inference Tasks in Bayesian Networks

Definition (Probabilistic Inference Task). Given random variables $X_1, \dots, X_n$, a **probabilistic inference task** consists of a set $\mathbf{X} \subseteq \{X_1, \dots, X_n\}$ of **query variables**, a set $\mathbf{E} \subseteq \{X_1, \dots, X_n\}$ of **evidence variables**, and an **event** $\mathbf{e}$ that assigns values to $\mathbf{E}$. We wish to compute the **posterior probability distribution** $\mathbf{P}(\mathbf{X} \mid \mathbf{e})$.

Notes:

- $\mathbf{Y} := \{X_1, \dots, X_n\} \setminus (\mathbf{X} \cup \mathbf{E})$ are the **hidden variables**.
- We assume that a *BN* for $X_1, \dots, X_n$ is given.
- In the remainder, for simplicity, $\mathbf{X} = \{\mathbf{X}\}$ is a singleton.

Example: In $\mathbf{P}(\textit{Burglary} \mid \textit{johncalls}, \textit{marycalls})$, $\mathbf{X} = \textit{Burglary}$, $\mathbf{e} = \textit{johncalls}, \textit{marycalls}$, and $\mathbf{Y} = \{\textit{Alarm}, \textit{EarthQuake}\}$.

Simplifying the Problem by Using Normalization

According to the definition of conditional probability:

$$P(X \mid \mathbf{E} = \mathbf{e}) = \frac{P(X, \mathbf{E} = \mathbf{e})}{P(\mathbf{E} = \mathbf{e})}$$

It is sufficient to compute for each $x \in D_X$ the value

$$P(X = x, \mathbf{E} = \mathbf{e}).$$

Together with

$$P(\mathbf{E} = \mathbf{e}) = \sum_{x \in D_X} P(X = x, \mathbf{E} = \mathbf{e})$$

this gives the desired posterior distribution.

→ To predict the desired conditional probability $P(X \mid \mathbf{E} = \mathbf{e})$, it suffices for us to compute the probability of (non-atomic) events: $P(X, \mathbf{E} = \mathbf{e})$!

Notation: As $\alpha = \frac{1}{\sum_{x \in D_X} P(X=x, \mathbf{E}=\mathbf{e})}$ is easily derived from $P(X \mid \mathbf{E} = \mathbf{e})$, we simply write $P(X \mid \mathbf{E} = \mathbf{e}) = \alpha P(X, \mathbf{E} = \mathbf{e})$ instead of $\frac{P(X, \mathbf{E}=\mathbf{e})}{P(\mathbf{E}=\mathbf{e})}$.

Normalization: Idea

Problem: We know $P(\text{cavity} \wedge \text{toothache})$ but don't know $P(\text{toothache})$:

$$P(\text{cavity} \mid \text{toothache}) = \frac{P(\text{cavity} \wedge \text{toothache})}{P(\text{toothache})} = \frac{0.12}{P(\text{toothache})}$$

Step 1: Case distinction over the values of Cavity :

$$P(\neg \text{cavity} \mid \text{toothache}) = \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})} = \frac{0.08}{P(\text{toothache})}$$

Step 2: Assuming placeholder $\alpha := 1/P(\text{toothache})$:

$$P(\text{cavity} \mid \text{toothache}) = \alpha P(\text{cavity} \wedge \text{toothache}) = \alpha 0.12$$

$$P(\neg \text{cavity} \mid \text{toothache}) = \alpha P(\neg \text{cavity} \wedge \text{toothache}) = \alpha 0.08$$

Step 3: Fixing toothache to be true, view $P(\text{cavity} \wedge \text{toothache})$ vs. $P(\neg \text{cavity} \wedge \text{toothache})$ as the **relative weights of $P(\text{cavity})$ vs. $P(\neg \text{cavity})$ within toothache** . Then normalize their summed-up weight to 1:

$$1 = \alpha(0.12 + 0.08) \Rightarrow \alpha = 1/(0.12 + 0.08) = 1/0.2 = 5$$

→ α is the **normalization constant** scaling the sum of relative weights to 1.

Normalization: Formal

Definition. Given a vector $\langle w_1, \dots, w_k \rangle$ of numbers in $[0, 1]$ where $\sum_{i=1}^k w_i \leq 1$, the **normalization constant** α is $\alpha \langle w_1, \dots, w_k \rangle := 1 / \sum_{i=1}^k w_i$.

Example: $\alpha \langle 0.12, 0.08 \rangle = 5 \langle 0.12, 0.08 \rangle = \langle 0.6, 0.4 \rangle$.

Proposition (Normalization). Given a random variable X and an event e , we have $P(X \mid e) = \alpha P(X, e)$.

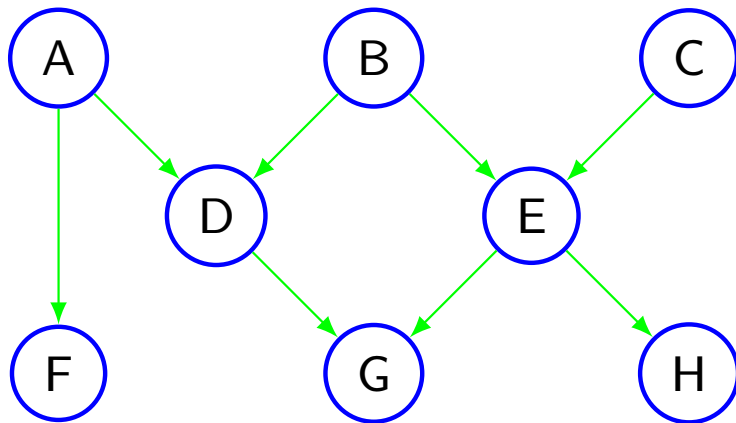
Proof. For each value x of X , $P(X = x \mid e) = P(X = x \wedge e) / P(e)$. So all we need to prove is that $\alpha = 1 / P(e)$. By definition, $\alpha = 1 / \sum_x P(X = x \wedge e)$, so we need to prove $P(e) = \sum_x P(X = x \wedge e)$ which holds by Marginalization.

Example: $\alpha \langle P(\text{cavity} \wedge \text{toothache}), P(\neg \text{cavity} \wedge \text{toothache}) \rangle = \alpha \langle 0.12, 0.08 \rangle$, so $P(\text{cavity} \mid \text{toothache}) = 0.6$, and $P(\neg \text{cavity} \mid \text{toothache}) = 0.4$.

Normalization+Marginalization: Given “query variable” X , “observed event” e , and “hidden variables” set $\mathbf{Y}$: $P(X \mid e) = \alpha P(X, e) = \alpha \sum_{\mathbf{y} \in \mathbf{Y}} P(X, e, \mathbf{y})$.

→ **Second of the four basic techniques in Bayesian networks.**

Naive Enumeration Example



$$\text{Find } P(B|a, f, g, h) = \frac{P(B, a, f, g, h)}{P(a, f, g, h)}$$

We can if we have access to $P(A, B, C, D, E, F, G, H)$

→ **Reminder: We can recover the full joint probability distribution from our BN!**

Inserting evidence we get:

$$P(B \mid a, f, g, h) = \alpha \cdot \sum_{C, D, E} P(a, B, C, D, E, f, g, h)$$

and

$$\frac{1}{\alpha} = P(a, f, g, h) = \sum_B P(B, a, f, g, h)$$

Questionnaire

Question!

Say we know $P(\text{likeschappi} \wedge \text{dog}) = 0.32$ and $P(\neg \text{likeschappi} \wedge \text{dog}) = 0.08$. Can we compute $P(\text{likeschappi} \mid \text{dog})$?

(A): Yes.

(B): No.

→ Yes, because we can compute $P(\text{dog}) = P(\text{likeschappi} \wedge \text{dog}) + P(\neg \text{likeschappi} \wedge \text{dog})$, and thus $P(\text{likeschappi} \mid \text{dog}) = \frac{P(\text{likeschappi} \wedge \text{dog})}{P(\text{dog})}$.

→ In other words, we can use Normalization: $\mathbf{P}(X \mid \mathbf{e}) = \alpha \mathbf{P}(X, \mathbf{e})$.

Inserting *LikesChappi* for X and *dog* for $\mathbf{e}$, we get $\mathbf{P}(\text{LikesChappi} \mid \text{dog}) = \alpha \mathbf{P}(\text{LikesChappi}, \text{dog}) = \alpha \langle P(\text{likeschappi} \wedge \text{dog}), P(\neg \text{likeschappi} \wedge \text{dog}) \rangle = \alpha \langle 0.32, 0.08 \rangle$.

→ So what is $P(\text{likeschappi} \mid \text{dog})$? 0.8, because $\alpha = 1/P(\text{dog}) = 1/(0.32 + 0.08) = 2.5$.

Marginalization: Inference as Summation

Want to compute: $P(X = x, \mathbf{E} = \mathbf{e})$

Problem: We do not know the value of hidden variables $\mathbf{Y}$

We sum over all possible values of the hidden variables!

Let X be the variable of interest, $\mathbf{E}$ the evidence variables, and $\mathbf{Y} = Y_1, \dots, Y_l$ the remaining variables in the network not belonging to $X \cup \mathbf{E}$. Then

$$P(X = x, \mathbf{E} = \mathbf{e}) = \sum_{y_1 \in D_{Y_1}} \dots \sum_{y_l \in D_{Y_l}} P(X = x, \mathbf{E} = \mathbf{e}, Y_1 = y_1, \dots, Y_l = y_l).$$

Note:

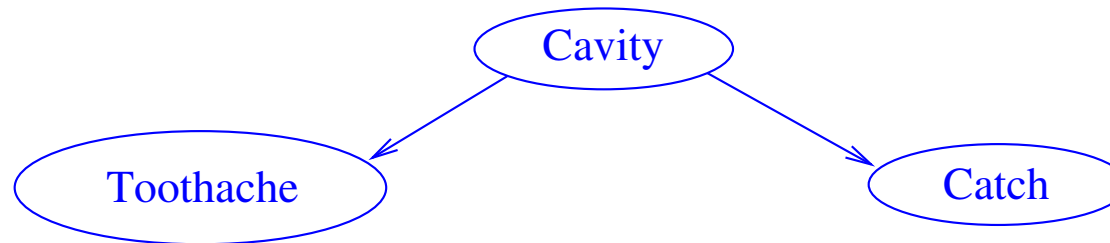
- For each $\mathbf{y}$ the probability $P(X = x, \mathbf{E} = \mathbf{e}, \mathbf{Y} = \mathbf{y})$ can be computed from the network (in time linear in the number of random variables).
- The number of configurations over $\mathbf{Y}$ is exponential in l .

Simplified notation:

$$P(x, \mathbf{e}) = \sum_{Y_1} \dots \sum_{Y_l} P(x, \mathbf{e}, Y_1, \dots, Y_l).$$

Exploiting Conditional Independence: Overview

1. Graph captures variable dependencies: (Variables $X_1, \dots, X_n$)



→ **Given evidence e , want to know $P(X | e)$.** Remaining vars: $\mathbf{Y}$.

2. Normalization+Marginalization:

$$P(\mathbf{X} | \mathbf{e}) = \alpha P(\mathbf{X}, \mathbf{e}); \text{ if } \mathbf{Y} \neq \emptyset \text{ then } P(\mathbf{X} | \mathbf{e}) = \alpha \sum_{\mathbf{y} \in \mathbf{Y}} P(\mathbf{X}, \mathbf{e}, \mathbf{y})$$

→ A sum over atomic events!

3. Chain rule: Order $X_1, \dots, X_n$ consistently with dependency graph.

$$P(X_1, \dots, X_n) = P(X_n | X_{n-1}, \dots, X_1) * P(X_{n-1} | X_{n-2}, \dots, X_1) * \dots * P(X_1)$$

4. **Exploit conditional independence:** Instead of $P(X_i | X_{i-1}, \dots, X_1)$, with previous slide we can use $P(\mathbf{X}_i | \text{Parents}(\mathbf{X}_i))$.

Exploiting Conditional Independence: Example

1. **Graph captures variable dependencies:** (See previous slide.)

→ **Given** *toothache, catch*, **want** $\mathbf{P}(Cavity \mid toothache, catch)$. Remaining vars: $\emptyset$.

2. **Normalization+Marginalization:**

$$\mathbf{P}(Cavity \mid toothache, catch) = \alpha \mathbf{P}(Cavity, toothache, catch)$$

3. **Chain rule:** Order $X_1 = Cavity$, $X_2 = Toothache$, $X_3 = Catch$.

$$\begin{aligned} &\mathbf{P}(Cavity, toothache, catch) = \\ &\mathbf{P}(catch \mid toothache, Cavity) \mathbf{P}(toothache \mid Cavity) \mathbf{P}(Cavity) \end{aligned}$$

4. **Exploit conditional independence:**

Instead of $\mathbf{P}(catch \mid toothache, Cavity)$ use $\mathbf{P}(catch \mid Cavity)$.

Thus: $\mathbf{P}(Cavity \mid toothache, catch) =$

$$\alpha \mathbf{P}(catch \mid Cavity) \mathbf{P}(toothache \mid Cavity) \mathbf{P}(Cavity) =$$

$$\alpha \langle 0.9 * 0.6 * 0.2, 0.2 * 0.1 * 0.8 \rangle = \alpha \langle 0.108, 0.016 \rangle. \text{ So } \alpha \approx 8.06 \text{ and}$$

$$\mathbf{P}(cavity \mid toothache \wedge catch) \approx 0.87.$$

Inference by Enumeration: John and Mary

- Want: $\mathbf{P}(\text{Burglary} \mid \text{johncalls}, \text{marycalls})$.
Hidden variables: $\mathbf{Y} = \{\text{Earthquake}, \text{Alarm}\}$.

- Normalization+Marginalization:**

$$\begin{aligned} \mathbf{P}(B \mid j, m) &= \alpha \mathbf{P}(B, j, m) \\ &= \alpha \sum_{v_E} \sum_{v_A} \mathbf{P}(B, j, m, v_E, v_A) \end{aligned}$$

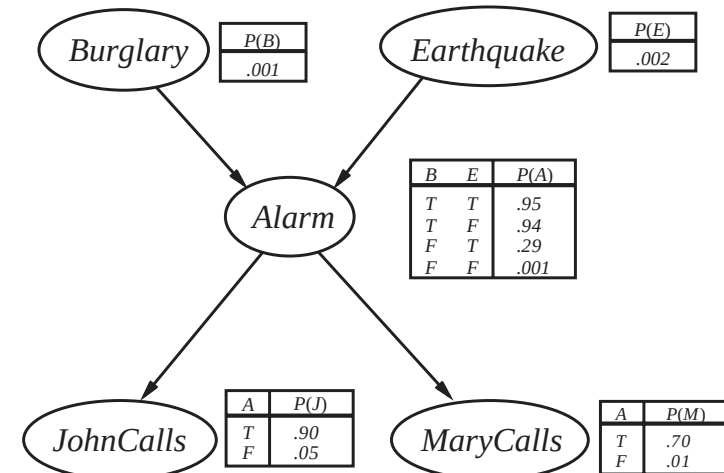
- Order** $X_1 = B, X_2 = E, X_3 = A, X_4 = J, X_5 = M$.

- Chain rule and conditional independence:** $\mathbf{P}(B \mid j, m) = \alpha \sum_{v_E} \sum_{v_A} \mathbf{P}(B)P(v_E)\mathbf{P}(v_A \mid B, v_E)P(j \mid v_A)P(m \mid v_A)$

- Move variables outwards** (until we hit the first parent): $\mathbf{P}(B \mid j, m) = \alpha \mathbf{P}(B) \sum_{v_E} P(v_E) \sum_{v_A} \mathbf{P}(v_A \mid B, v_E)P(j \mid v_A)P(m \mid v_A)$

→ The probabilities of the outside-variables multiply the entire “rest of the sum”.

- Continuation on next slide ...



Inference by Enumeration: John and Mary, ctd.

Chain rule and conditional independence, ctd.: $\mathbf{P}(B \mid j, m) =$

$$\alpha \mathbf{P}(B) \sum_{v_E} P(v_E) \sum_{v_A} \mathbf{P}(v_A \mid B, v_E) P(j \mid v_A) P(m \mid v_A)$$

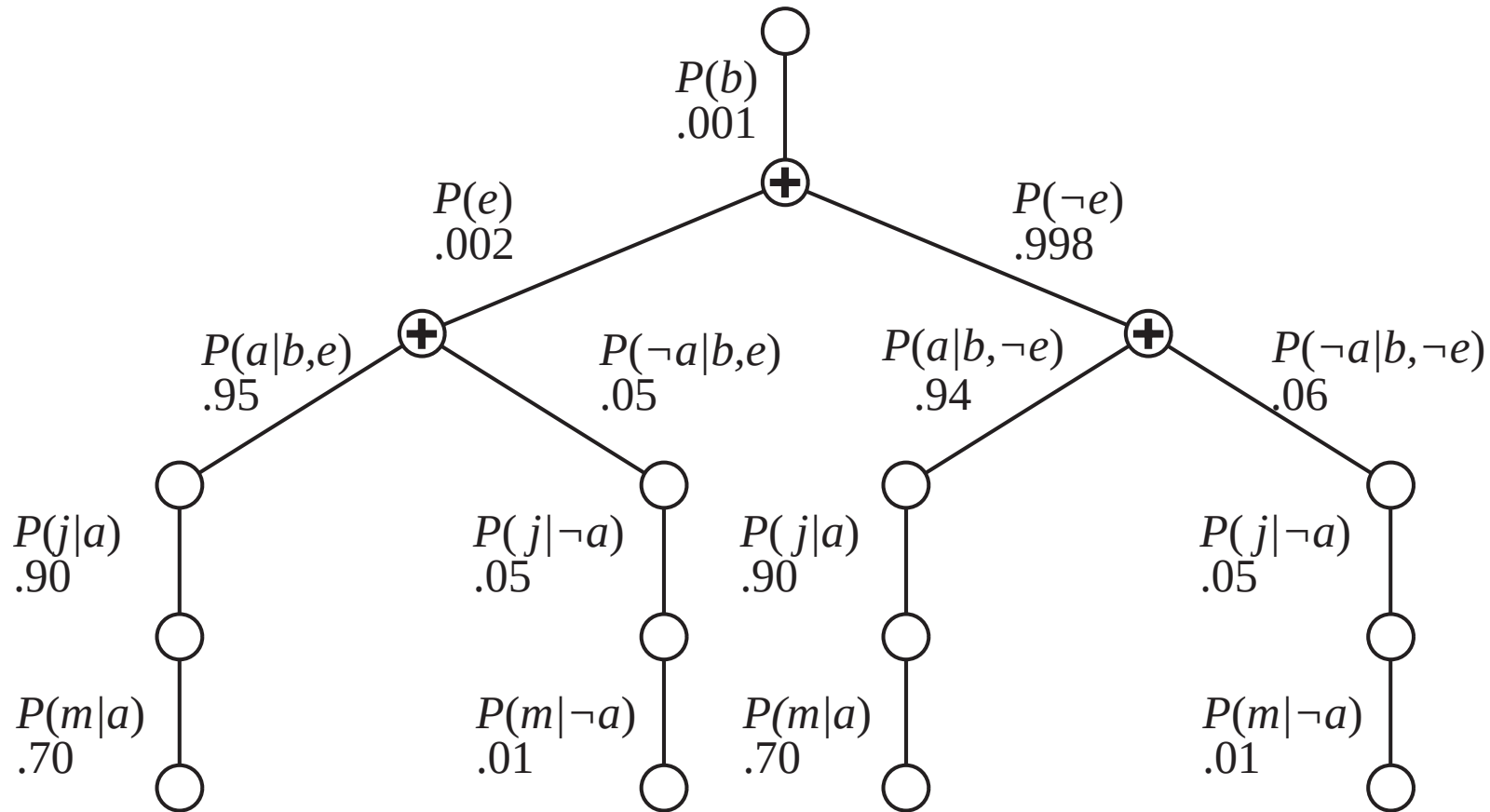
$$\alpha \langle P(b) [P(e) \underbrace{(P(a \mid b, e) P(j \mid a) P(m \mid a) + P(\neg a \mid b, e) P(j \mid \neg a) P(m \mid \neg a))}_e + P(\neg e) \underbrace{(P(a \mid b, \neg e) P(j \mid a) P(m \mid a) + P(\neg a \mid b, \neg e) P(j \mid \neg a) P(m \mid \neg a))}_{\neg e}] ,$$

$$P(\neg b) [P(e) \underbrace{(P(a \mid \neg b, e) P(j \mid a) P(m \mid a) + P(\neg a \mid \neg b, e) P(j \mid \neg a) P(m \mid \neg a))}_e + P(\neg e) \underbrace{(P(a \mid \neg b, \neg e) P(j \mid a) P(m \mid a) + P(\neg a \mid \neg b, \neg e) P(j \mid \neg a) P(m \mid \neg a))}_{\neg e}] \rangle$$

$$= \alpha \langle 0.00059224, 0.0014919 \rangle \approx \langle 0.284, 0.716 \rangle$$

→ This computation can be viewed as a “search tree”, see next slide.

The Evaluation of $P(b | j, m)$, as a “Search Tree”



→ Inference by enumeration = a tree with “sum nodes” branching over values of hidden variables, and with non-branching “multiplication nodes”.

Inference by Enumeration: Properties

Inference by Enumeration:

- Evaluates the tree in a depth-first manner.
- **Space Complexity:** Linear in the number of variables.
- **Time Complexity:** Exponential in the number of hidden variables, e.g., $O(2^{|\mathbf{Y}|})$ in case these variables are Boolean.
→ **Can we do better than this?**

Bad News: Not in general.

- Probabilistic inference is **#P**-hard.
- **#P** is harder than **NP** (i.e., $\mathbf{NP} \subseteq \mathbf{\#P}$).

But: **Variable Elimination.**

- Improves on inference by enumeration through **(A) avoiding repeated computation**, and **(B) avoiding irrelevant computation**.
- In some special cases, variable elimination runs in polynomial time.

Variable Elimination: Sketch of Ideas

(A) Avoiding repeated computation: Evaluate expressions from right to left, storing all intermediate results. For query $P(B \mid j, m)$:

① CPTs of BN yield **factors** (probability tables): $P(B \mid j, m) =$

$$\alpha \underbrace{P(B)}_{f_1(B)} \sum_{v_E} \underbrace{P(v_E)}_{f_2(E)} \sum_{v_A} \underbrace{P(v_A \mid B, v_E)}_{f_3(A, B, E)} \underbrace{P(j \mid v_A)}_{f_4(A)} \underbrace{P(m \mid v_A)}_{f_5(A)}$$

② Then the computation is performed in terms of **factor product** and **summing out variables** from factors: $P(B \mid j, m) =$

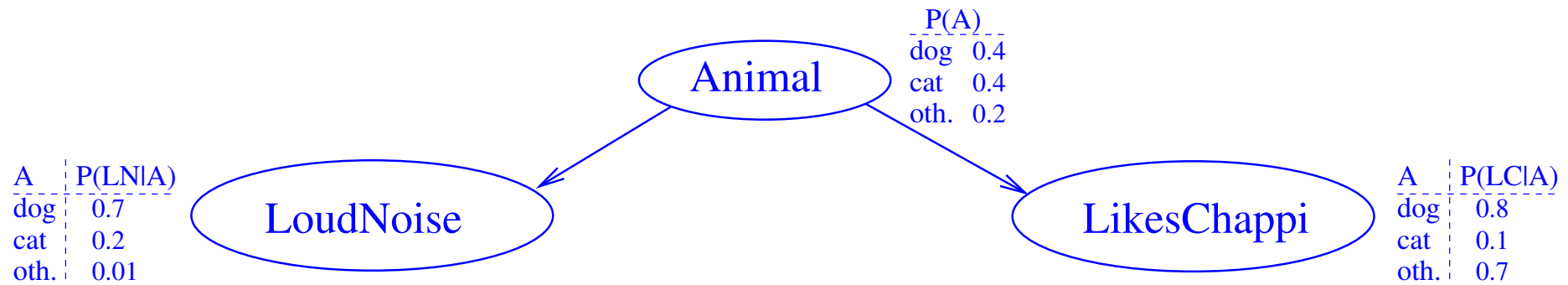
$$\alpha f_1(B) \times \sum_{v_E} f_2(E) \times \sum_{v_A} f_3(A, B, E) \times f_4(A) \times f_5(A)$$

(B) Avoiding irrelevant computation: Repeatedly remove hidden variables that are leaf nodes. For query $P(\text{JohnCalls} \mid \text{burglary})$:

$$P(J \mid b) = \alpha P(b) \sum_{v_E} P(v_E) \sum_{v_A} P(v_A \mid b, v_E) P(J \mid v_A) \sum_{v_M} P(v_M \mid v_A)$$

→ The rightmost sum equals 1 and can be dropped.

Questionnaire



Question!

Say BN is the Bayesian network above. How can we compute $P(dog \mid loudnoise)$?

$P(d \mid ln) =$

(A): $\alpha \sum_{v_{LC}} P(v_{LC} \mid d) P(ln \mid d)$

(B): $\alpha \sum_{v_{LC}} P(v_{LC} \mid d) P(ln \mid d) P(d)$

(C): $\alpha P(ln \mid d) P(d) \sum_{v_{LC}} P(v_{LC} \mid d)$

(D): $\alpha P(ln \mid d) P(d)$

(A) No: need to multiply by $P(d)$.

(B) Yes.

(C) Yes: $P(ln \mid d)$ and $P(d)$ do not depend on lc .

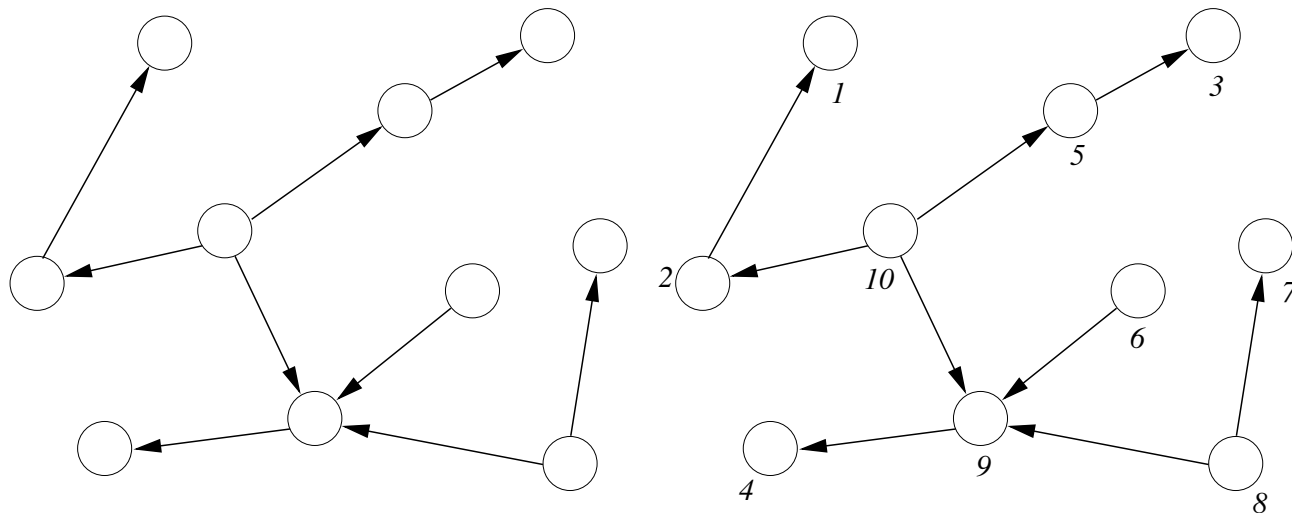
(D) Yes: $\sum_{v_{LC}} P(v_{LC} \mid d) = 1$.

→ So what is $P(dog \mid loudnoise)$? We compute $\alpha \langle P(ln \mid d) P(d), P(ln \mid c) P(c), P(ln \mid o) P(o) \rangle = \alpha \langle 0.7 * 0.4, 0.2 * 0.4, 0.01 * 0.2 \rangle \approx \langle 0.77, 0.22, 0.01 \rangle$. Hence $P(dog \mid loudnoise) = 0.77$. Which BTW is $> P(dog \mid likeschappi) = 0.64$.

Variable Elimination Runtime

An important easy case:

- A graph is called **singly connected**, or a **polytree**, if there is at most one undirected path between any two nodes in the graph.



→ Is our BN for Mary & John a polytree? Yes.

For singly connected network: any elimination order that “peels” variables from outside will only create factors of only one variable.

The complexity of inference is therefore **linear in the total size of the network** (= combined size of all conditional probability tables).

D-separation

Given a **Causal Network** that relates causes to effects, what information is relevant to predict the value of a variable?

In the context of Bayesian Networks:

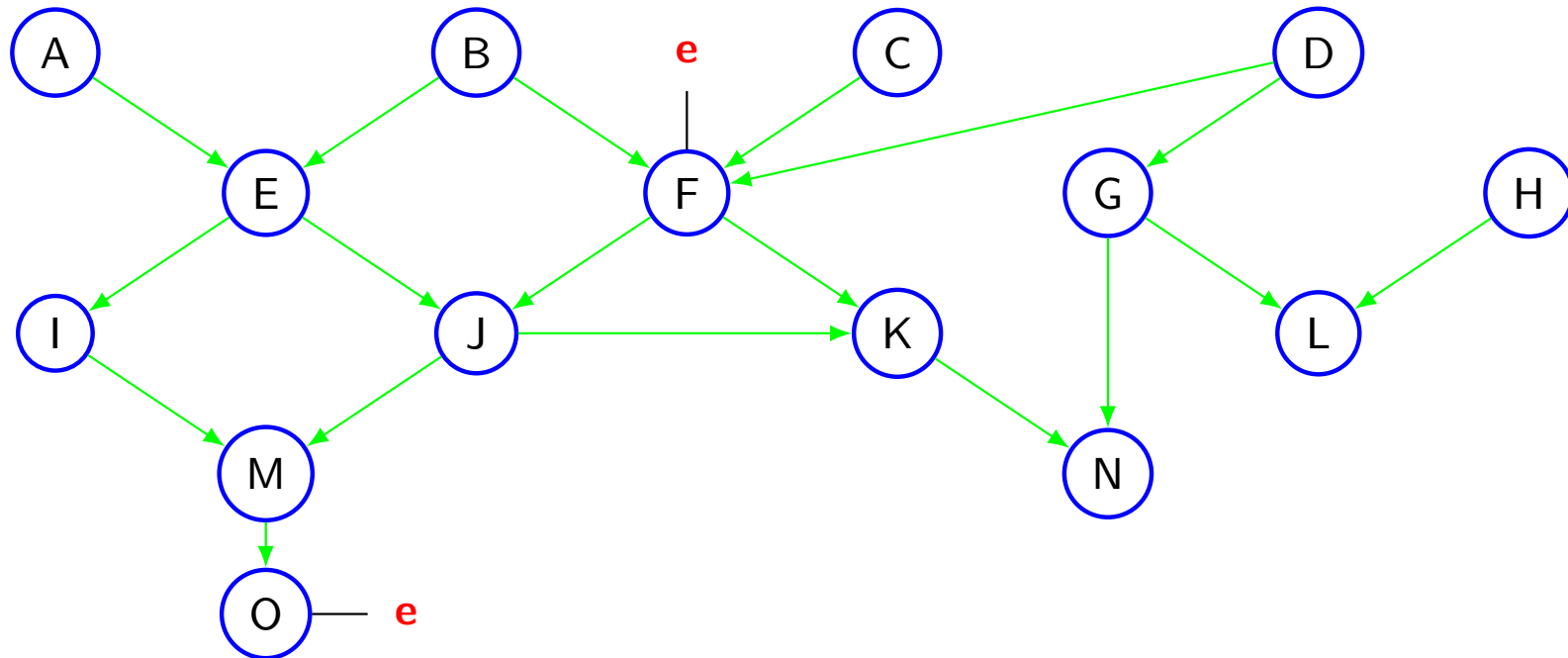
Let X , Y , and E be disjoint sets of variables.

Is X conditionally independent from Y given E ?

→ We introduce d-separation as a general way of answering that question

Note: Even if X and Y are independent that does not mean that they are conditionally independent given E . For example, two dice rolls are independent, but if we have evidence that the sum is 10, then they are not independent anymore.

Transmission of Evidence: Questionnaire



Question!

Given evidence **e**, can knowledge of *A* have an impact on our knowledge of _?

(A): *J*

(B): *G*

(C): *H*

(D): *B*

(A) yes

(B) yes

(C) no

(D) yes (note O is a descendant of E)

Bayesian Network Example 1



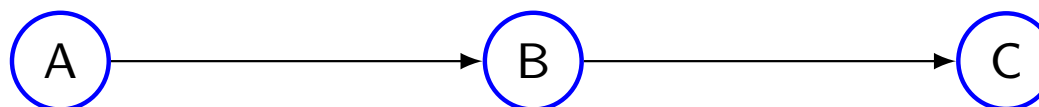
- If there has been a flooding does that tell me something about the amount of rain that has fallen?

Yes, Rainfall and Flooding are not independent!

- The water level is high: If there has been a flooding does that tell me anything new about the amount of rain that has fallen?

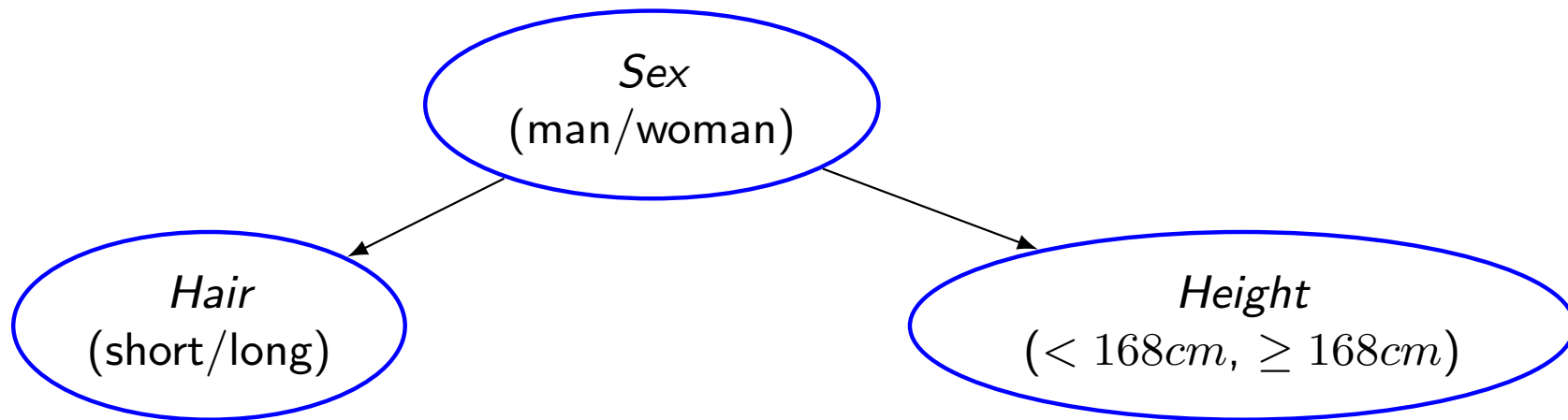
No, Rainfall and Flooding are conditionally independent given WaterLevel!

Serial Connection (Indirect cause)



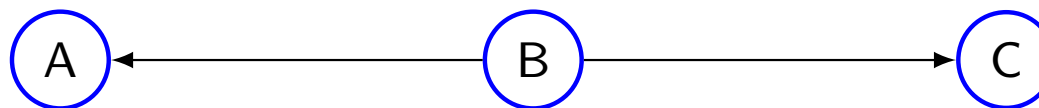
Information can be transmitted between A and C through B if B is not observed. If B is observed, A and C are independent.

Bayesian Network Example 2



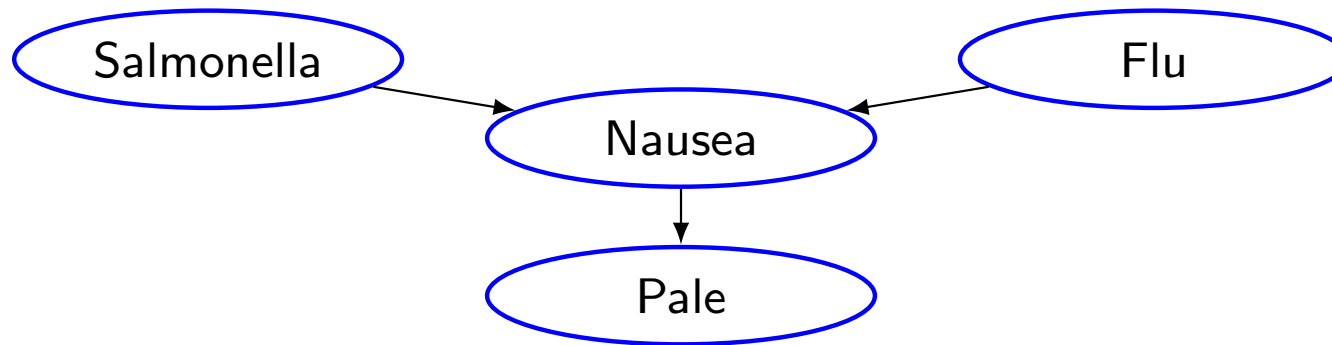
- If a person has long hair does that say something about his/her stature?
Yes, Hair and Height are not independent!
- It is a woman: If she has long hair does that say something about her stature?
No, Hair and Height are conditionally independent given Sex!

Diverging Connection (common cause)



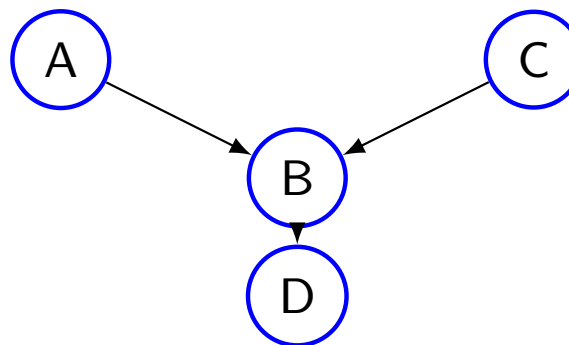
Information can be transmitted between A and C through B if B is not observed. If B is observed, A and C are independent.

Bayesian Network Example 3



- Does salmonella have an impact on Flu?
No, salmonella is independent of Flu
- If a person is Pale, does salmonella then have an impact on Flu?
Yes, salmonella can explain why the person is pale!

Converging Connection (common effect)



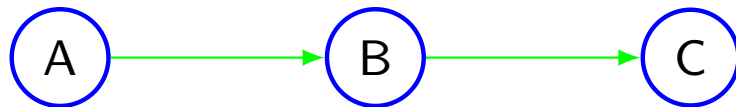
A and C are independent unless we have information about B or its descendants.

If B or a descendant (D) are observed, we can transmit information between A and C .

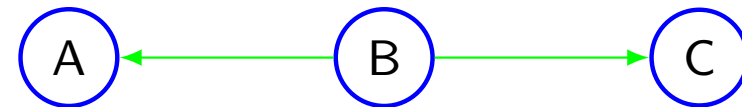
D-Separation Rules

Does evidence from A affect our knowledge about C?

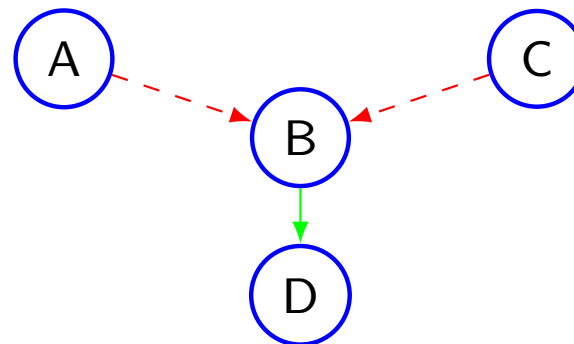
- Evidence may be transmitted through a serial or diverging connection unless it is instantiated.
- Evidence may be transmitted through a converging connection only if either the variable in the connection or one of its descendants has received evidence.



Serial



Diverging

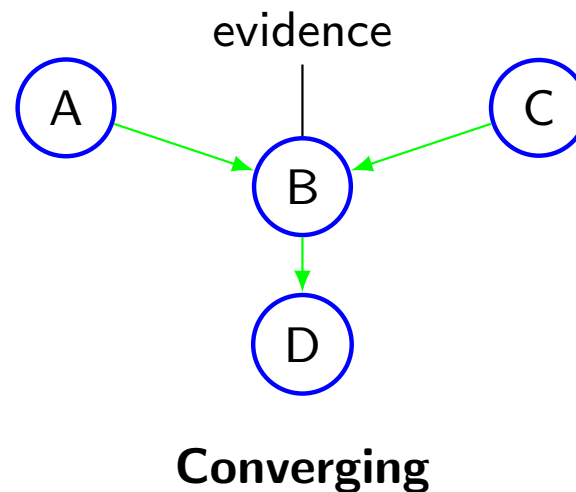
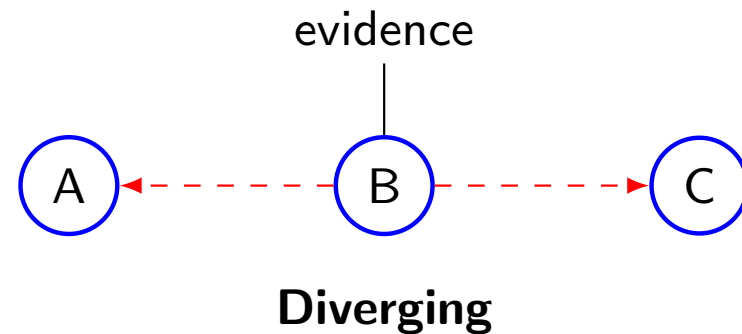
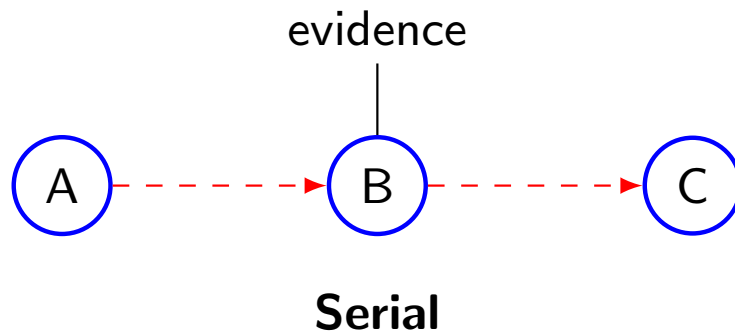


Converging

D-Separation Rules

Does evidence from A affect our knowledge about C?

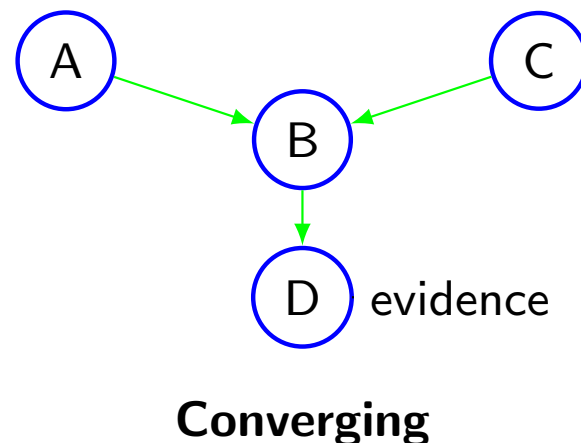
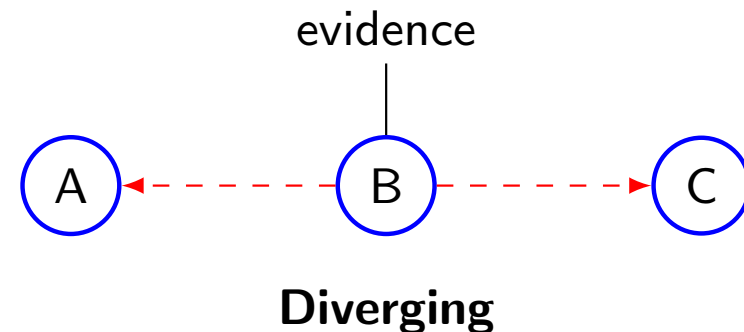
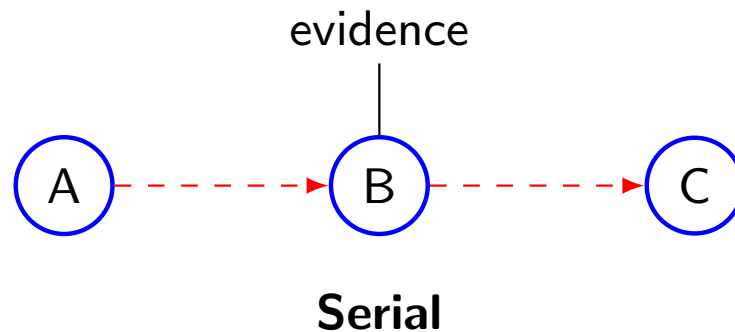
- Evidence may be transmitted through a serial or diverging connection unless it is instantiated.
- Evidence may be transmitted through a converging connection only if either the variable in the connection or one of its descendants has received evidence.



D-Separation Rules

Does evidence from A affect our knowledge about C?

- Evidence may be transmitted through a serial or diverging connection unless it is instantiated.
- Evidence may be transmitted through a converging connection only if either the variable in the connection or one of its descendants has received evidence.



The d-separation theorem

Theorem

For all pairwise disjoint sets $\mathbf{A}, \mathbf{B}, \mathbf{C}$ of nodes in a Bayesian network:

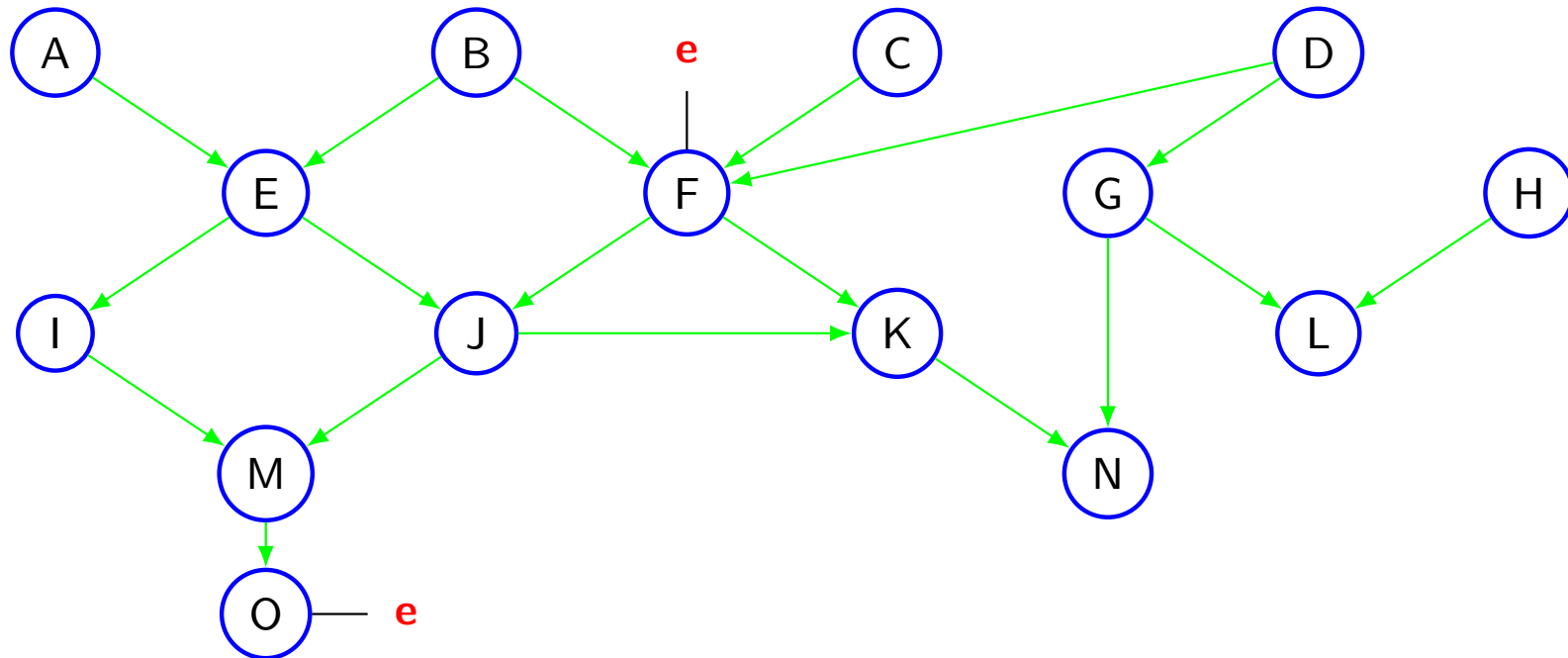
If $\mathbf{C}$ d-separates $\mathbf{A}$ from $\mathbf{B}$, then $P(\mathbf{A} \mid \mathbf{B}, \mathbf{C}) = P(\mathbf{A} \mid \mathbf{C})$.

There are no more general graphical conditions than d-separation for which such a result holds.

Why is d-separation important?

- Gaining insight: given a (correct) Bayesian network model, can derive insight into the dependencies among the variables
- Debugging a model: given a Bayesian network model, check whether entailed independence relations are plausible
- Correctness of algorithms: certain computational procedures depend on validity of special independence relations

Transmission of Evidence: Questionnaire



Question!

Given evidence **e**, can knowledge of *A* have an impact on our knowledge of _?

(A): *J*

(B): *G*

(C): *H*

(D): *B*

(A) yes

(B) yes

(C) no

(D) yes (note O is a descendant of E)

D-separation Algorithm

Algorithm for testing whether A and B are d-separated given evidence E

1. Construct the subgraph consisting of variables A , B and E plus all of their ancestors (for all nodes added, include also their parents).
2. Add undirected edges between all parents of a common child.
3. Replace directed edges by undirected edges
4. Remove all evidence E nodes from the graph (along with their connections).
5. If A and B are d-separated if there is no path between them in the resulting graph.

→ This algorithm basically tests whether there is a path from A to B while following the propagation rules described in the previous slides.

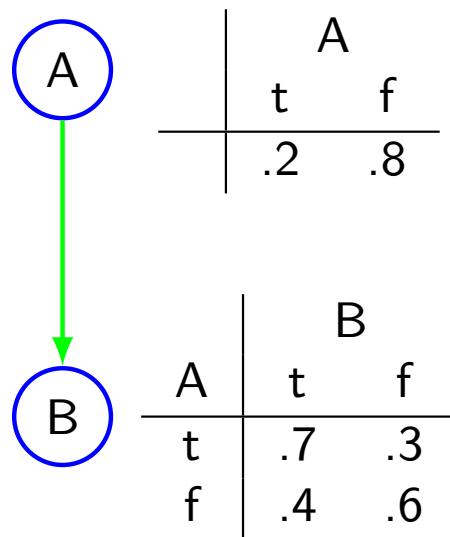
Sampling Inference

Using a BN as a Sample Generator

Observation: can use Bayesian network as random generator that produces states $\mathbf{X} = \mathbf{x}$ according to distribution P defined by the network.

→ We just sample a random value for each variable **always picking a value for parents(X) before picking a value for X**

Example:



- Generate random numbers r_A, r_B uniformly from $[0,1]$.
- Set $A = t$ if $r_A \leq .2$ and $A = f$ else.
- Depending on the value of A and r_B set B to t or f .

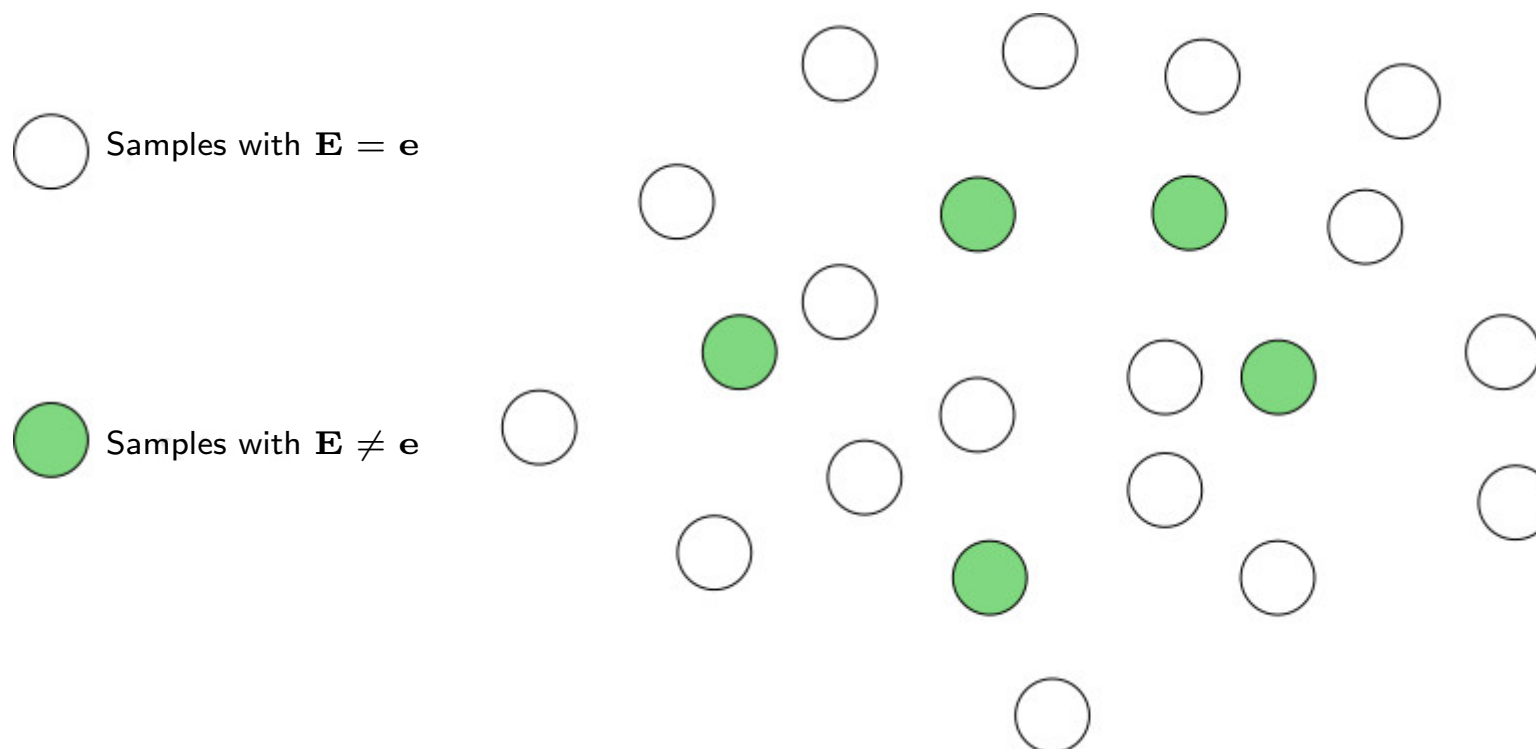
Random generation of one state: linear in size of network.

Sampling Inference

Approximate Inference from Samples

To compute an approximation of $P(\mathbf{E} = \mathbf{e})$ ($\mathbf{E}$ a subset of the variables in the Bayesian network):

- generate a (large) number of random states
- count the frequency of states in which $\mathbf{E} = \mathbf{e}$.



Accuracy

Hoeffding Bound

- p : true probability $P(\mathbf{E} = \mathbf{e})$
- s : estimate for p from sample of size n
- ϵ : an error bound > 0 .

Then

$$P(|s - p| > \epsilon) \leq 2e^{-2n\epsilon^2}$$

Required Sample Size

To obtain an estimate that with probability at most δ has an error greater than ϵ , it is sufficient to take

$$n = -\ln(\delta/2)/(2\epsilon^2) \text{ samples.}$$

Example

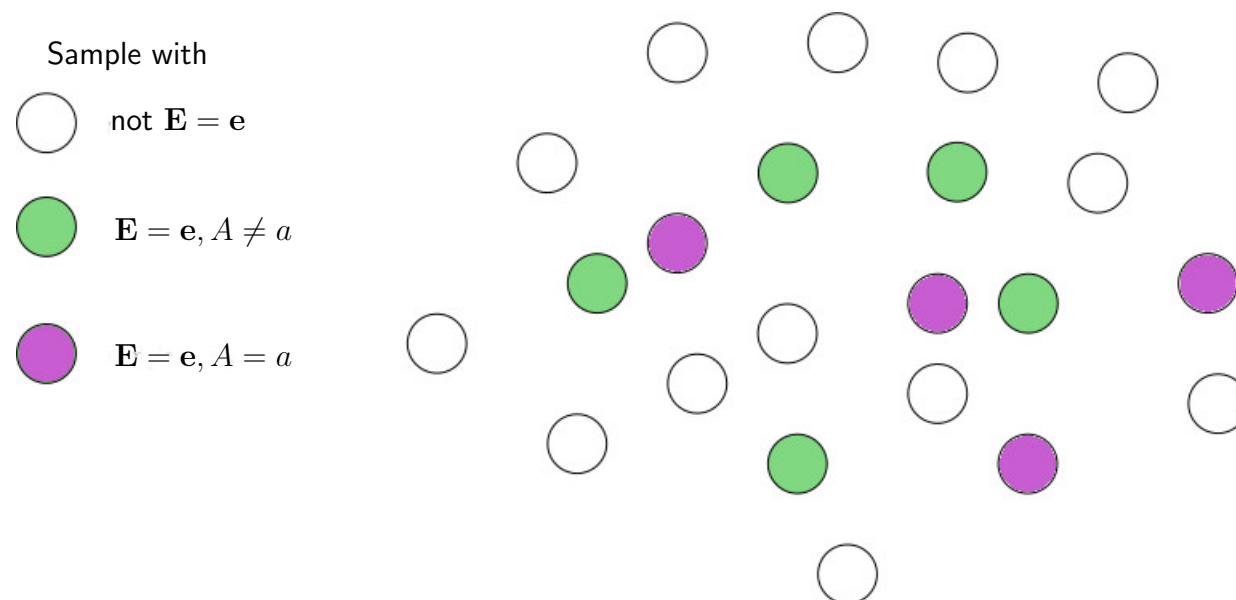
To get an error ϵ of less than 0.1 in 95% of the cases ($\delta = 0.05$), we need:

$$n > -\ln(0.05/2)/(2 \cdot 0.1^2) \approx 184 \text{ samples}$$

How many samples do we need if the error should be less than 0.01? 18444 samples

Rejection Sampling

The simplest approach: **Rejection Sampling**



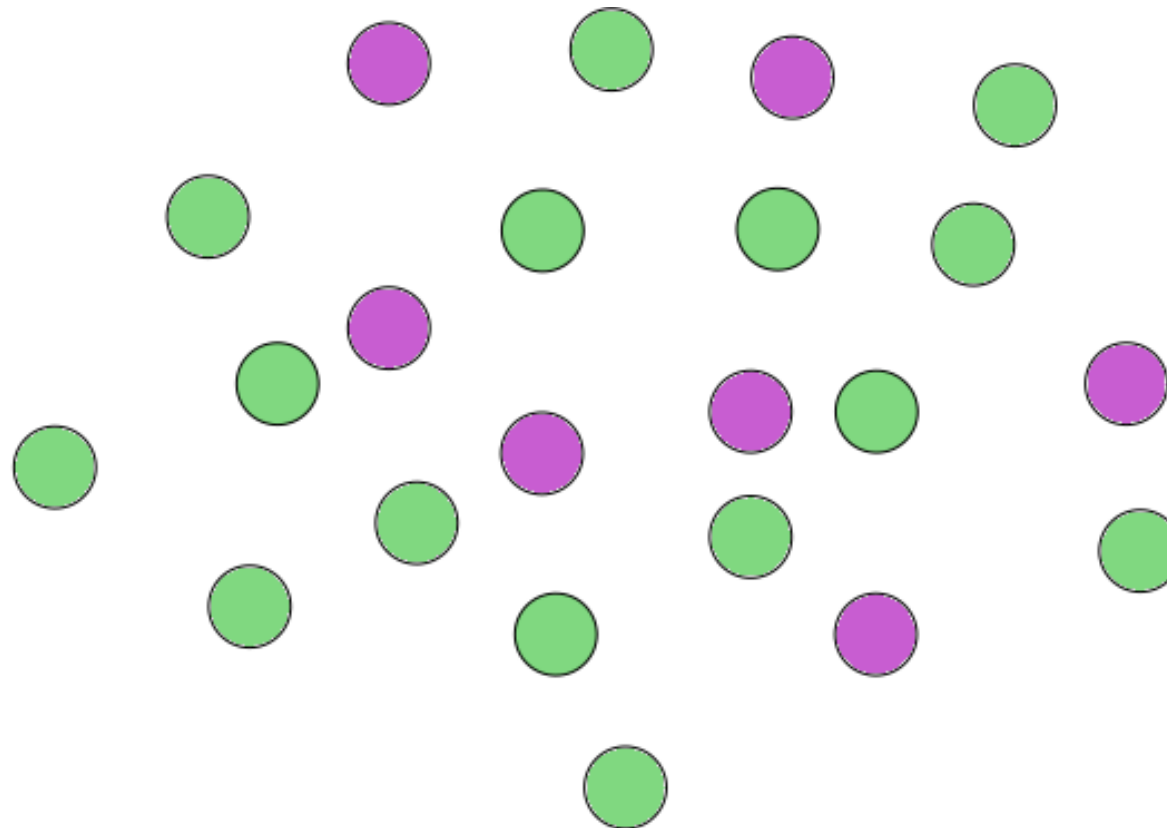
Approximation for $P(A = a \mid \mathbf{E} = \mathbf{e})$:

$$\frac{\# \text{ } \bullet}{\# \text{ } \bullet \cup \text{ } \bullet}$$

Sampling from the conditional distribution

Problem with rejection sampling: samples with $\mathbf{E} \neq \mathbf{e}$ are useless!

Ideally: would draw samples directly from the conditional distribution $P(\mathbf{A} \mid \mathbf{E} = \mathbf{e})$.



A Wrong Sampling Method

First idea (not to be followed)

- 1 Fix evidence variables to their observed states.
- 2 Sample from non-evidence variables.
- 3 Count frequency as before

Problem: This gives a sampling distribution

$$\prod_{X \in \mathbf{X} \setminus \mathbf{E}} P(X \mid \text{pa}(X) \setminus \mathbf{E}, \text{pa}(X) \cap \mathbf{E})$$

somewhere between $P(\mathbf{X})$ and $P(\mathbf{X} \mid \mathbf{e})$.

Likelihood Weighting

We would like to sample from

$$P(\mathbf{X}, \mathbf{e}) = \underbrace{\prod_{X \in \mathbf{X} \setminus \mathbf{E}} P(X \mid \text{pa}(X) \setminus \mathbf{E}, \text{pa}(X) \cap \mathbf{E})}_{\text{Part 1}} \cdot \underbrace{\prod_{E \in \mathbf{E}} P(E = e \mid \text{pa}(E) \setminus \mathbf{E}, \text{pa}(E) \cap \mathbf{E})}_{\text{Part 2}}$$

So instead weigh each generated sample with a weight corresponding to Part 2.

Estimate $P(X = x \mid \mathbf{e})$ as

$$\hat{P}(X = x \mid \mathbf{e}) = \frac{\sum_{\text{sample}: X=x} w(\text{sample})}{\sum_{\text{sample}} w(\text{sample})},$$

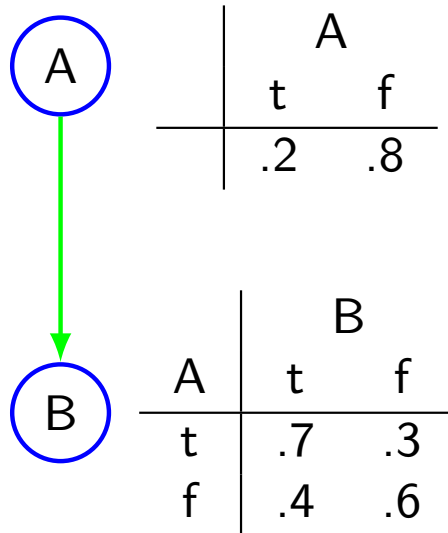
where

$$w(\text{sample}) = \prod_{E \in \mathbf{E}} P(E = e \mid \text{pa}(E) = \pi) \quad (\text{Part 2})$$

and π is the values of $\text{pa}(E)$ under *sample* and $\mathbf{e}$.

Likelihood Sampling: Example

Sample state where $B=f$



- Initialize weight $W = 1$
- Generate random number r_A uniformly from $[0,1]$.
- Set $A = t$ if $r_1 \leq .2$ and $A = f$ else.
- Set B to f . Update weight depending on the value of A to: $P(B = f \mid A = a)$

So:

- 20% of the time we sample $\langle A = t, B = f \rangle$ with weight 0.3
- and 80% of the time we sample $\langle A = f, B = f \rangle$ with weight 0.6

Summary

- **Bayesian networks (BN)** are a wide-spread tool to model uncertainty, and to reason about it. A BN represents **conditional independence relations** between random variables. It consists of a graph encoding the variable dependencies, and of **conditional probability tables (CPTs)**.
- **Probabilistic inference** requires to compute the probability distribution of a set of **query variables**, given a set of **evidence variables** whose values we know. The remaining variables are **hidden**.
- **Inference by enumeration** takes a BN as input, then applies **Normalization+Marginalization**, the **Chain rule**, and exploits conditional independence. This can be viewed as a tree search that branches over all values of the hidden variables.
- **Variable elimination** avoids unnecessary computation. It runs in polynomial time for poly-tree BNs. In general, exact probabilistic inference is **#P-hard**. Approximate probabilistic inference methods exist.
- When exact inference is infeasible, **approximate inference** can be used to obtain estimates faster.
- Approximate inference can be done via **sampling** from the distribution the BN represents. We can apply a weighting function to correct our estimations when we are interested in a different distribution.

Topics We Didn't Cover Here

- **More advance inference by sampling:** A whole zoo of methods for doing this exists.
- **Clustering:** Pre-combining subsets of variables to reduce the runtime of inference.
- **Compilation to SAT:** More precisely, to “weighted model counting” in CNF formulas. Model counting extends DPLL with the ability to determine the number of satisfying interpretations. Weighted model counting allows to define a mass for each such interpretation (= the probability of an atomic event).
- **Dynamic BN:** BN with one slice of variables at each “time step”, encoding probabilistic behavior over time.
- **Relational BN:** BN with predicates and object variables.
- **First-order BN:** Relational BN with quantification, i.e., probabilistic logic. E.g., the BLOG language developed by Stuart Russel and co-workers.

Reading

- *Chapter 8: Reasoning with Uncertainty* from the book “Artificial Intelligence: Foundations of Computational Agents” (2nd edition). In particular:
 - Section 8.4 “Probabilistic Inference”
 - Section 8.4.1 “Variable Elimination for Belief Networks”
 - Section 8.6 “Stochastic Simulation”
 - Section 8.6.5 “Importance Sampling”

For a further reading on the topic you can also read:

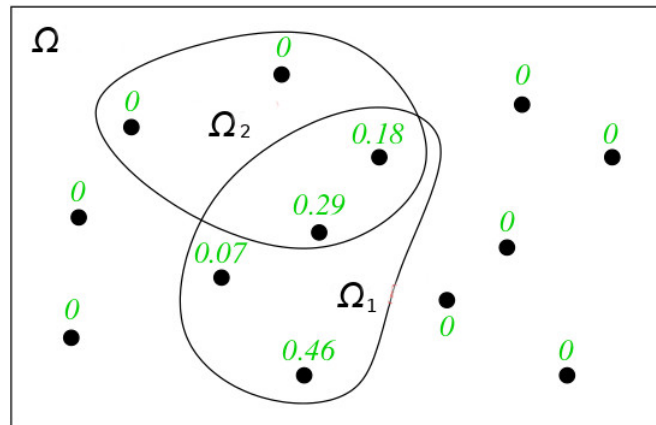
- *Chapter 14: Probabilistic Reasoning* from the book “Artificial Intelligence: A Modern Approach (4th edition)

Importance Sampling I

Importance sampling

Likelihood weighting is an instance of importance sampling, where

- samples are weighted and can come from (almost) any proposal distribution.
- $\mathbf{S}$: the set of all variables defining possible worlds (includes the variables A and $\mathbf{E}$).
- Possible worlds then are tuples s of values



$$P(A = a \mid \mathbf{E} = \mathbf{e}) = \sum_{s \in \Omega} P(A = a \mid \mathbf{S} = s) P(\mathbf{S} = s \mid \mathbf{E} = \mathbf{e})$$

Observe that:

- $P(\mathbf{S} = s \mid \mathbf{E} = \mathbf{e})$ are the "green numbers"
- $P(A = a \mid \mathbf{S} = s)$ is 0 or 1 depending on whether $A = a$ in s .

Importance Sampling II

If $s_1, \dots, s_n$ are sampled according to $P(\mathbf{S} = \mathbf{s} \mid \mathbf{E} = \mathbf{e})$, then

$$P(A = a \mid \mathbf{E} = \mathbf{e}) = \sum_{\mathbf{s} \in \Omega} P(A = a \mid \mathbf{S} = \mathbf{s}) P(\mathbf{S} = \mathbf{s} \mid \mathbf{E} = \mathbf{e}) \approx \frac{1}{n} \sum_{i=1}^n P(A = a \mid \mathbf{S} = \mathbf{s}_i)$$

Let Q be **any** probability distribution according to which we can sample possible worlds $\mathbf{s}_i$ (called a **proposal distribution**). Then:

$$\sum_{\mathbf{s} \in \Omega} P(A = a \mid \mathbf{S} = \mathbf{s}) P(\mathbf{S} = \mathbf{s} \mid \mathbf{E} = \mathbf{e}) = \sum_{\mathbf{s} \in \Omega} P(A = a \mid \mathbf{S} = \mathbf{s}) \frac{P(\mathbf{S} = \mathbf{s} \mid \mathbf{E} = \mathbf{e})}{Q(\mathbf{S} = \mathbf{s})} Q(\mathbf{S} = \mathbf{s})$$

If $s_1, \dots, s_n$ are sampled according to Q , then the right side is approximated by

$$\frac{1}{n} \sum_{i=1}^n P(A = a \mid \mathbf{S} = \mathbf{s}_i) \frac{P(\mathbf{S} = \mathbf{s}_i \mid \mathbf{E} = \mathbf{e})}{Q(\mathbf{S} = \mathbf{s}_i)}$$

which then is also an approximation of $P(A = a \mid \mathbf{E} = \mathbf{e})$.

Importance Sampling III

Importance Sampling

- Generate random samples $\mathbf{s}_i$ according to some proposal distribution Q .
- Estimate $P(A = a \mid \mathbf{E} = \mathbf{e})$ by $\frac{1}{n} \sum_{i=1}^n P(A = a \mid \mathbf{S} = \mathbf{s}_i) \frac{P(\mathbf{S}=\mathbf{s}_i \mid \mathbf{E}=\mathbf{e})}{Q(\mathbf{S}=\mathbf{s}_i)}$

Observations and Issues

- $P(A = a \mid \mathbf{S} = \mathbf{s}_i)$ is still only 0-1-valued
- $P(\mathbf{S} = \mathbf{s}_i \mid \mathbf{E} = \mathbf{e})$ is usually easy to compute, because $\mathbf{s}_i$ contains a value for all variables.
- $P(\mathbf{S} = \mathbf{s}_i \mid \mathbf{E} = \mathbf{e}) = 0$ if $\mathbf{s}_i$ does not satisfy $\mathbf{E} = \mathbf{e}$, i.e. samples that do not comply with the evidence don't count.
- The best approximation is obtained when Q is close to (identical to) $P(\mathbf{S} \mid \mathbf{E} = \mathbf{e})$.

Variable Elimination: Sketch of Ideas

(A) Avoiding repeated computation: Evaluate expressions from right to left, storing all intermediate results. For query $P(B \mid j, m)$:

① CPTs of BN yield **factors** (probability tables): $P(B \mid j, m) =$

$$\alpha \underbrace{P(B)}_{f_1(B)} \sum_{v_E} \underbrace{P(v_E)}_{f_2(E)} \sum_{v_A} \underbrace{P(v_A \mid B, v_E)}_{f_3(A, B, E)} \underbrace{P(j \mid v_A)}_{f_4(A)} \underbrace{P(m \mid v_A)}_{f_5(A)}$$

② Then the computation is performed in terms of **factor product** and **summing out variables** from factors: $P(B \mid j, m) =$

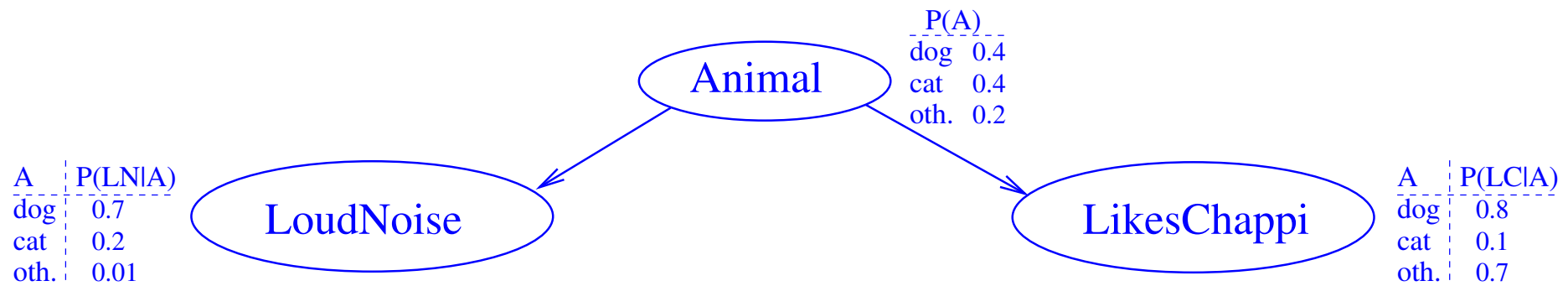
$$\alpha f_1(B) \times \sum_{v_E} f_2(E) \times \sum_{v_A} f_3(A, B, E) \times f_4(A) \times f_5(A)$$

(B) Avoiding irrelevant computation: Repeatedly remove hidden variables that are leaf nodes. For query $P(\text{JohnCalls} \mid \text{burglary})$:

$$P(J \mid b) = \alpha P(b) \sum_{v_E} P(v_E) \sum_{v_A} P(v_A \mid b, v_E) P(J \mid v_A) \sum_{v_M} P(v_M \mid v_A)$$

→ The rightmost sum equals 1 and can be dropped.

Questionnaire



Question!

Say BN is the Bayesian network above. How can we compute $P(dog \mid loudnoise)$?

$P(d \mid ln) =$

(A): $\alpha \sum_{v_{LC}} P(v_{LC} \mid d) P(ln \mid d)$

(B): $\alpha \sum_{v_{LC}} P(v_{LC} \mid d) P(ln \mid d) P(d)$

(C): $\alpha P(ln \mid d) P(d) \sum_{v_{LC}} P(v_{LC} \mid d)$

(D): $\alpha P(ln \mid d) P(d)$

(A) No: need to multiply by $P(d)$.

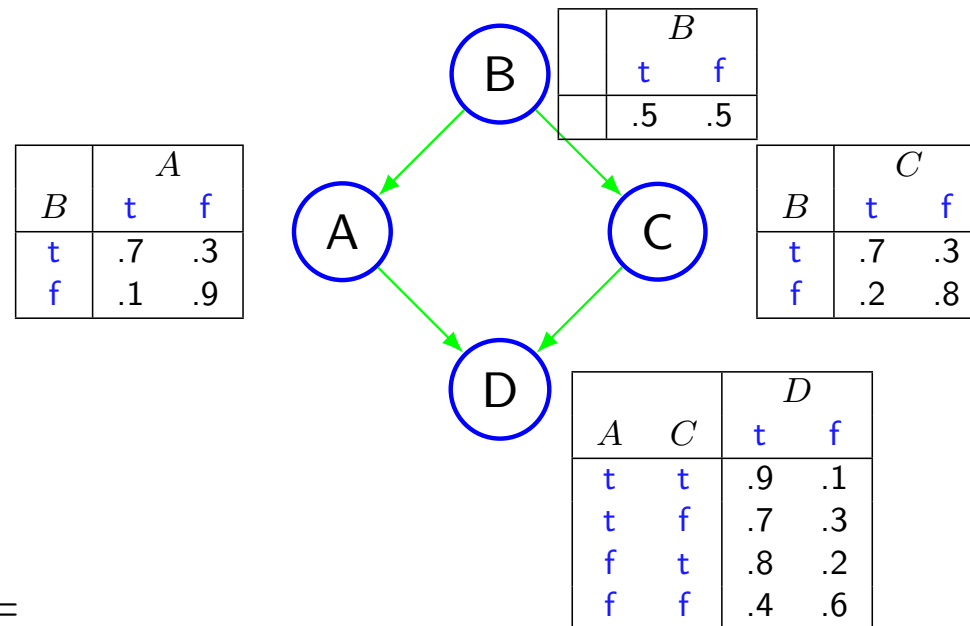
(B) Yes.

(C) Yes: $P(ln \mid d)$ and $P(d)$ do not depend on lc .

(D) Yes: $\sum_{v_{LC}} P(v_{LC} \mid d) = 1$.

→ So what is $P(dog \mid loudnoise)$? We compute $\alpha \langle P(ln \mid d) P(d), P(ln \mid c) P(c), P(ln \mid o) P(o) \rangle = \alpha \langle 0.7 * 0.4, 0.2 * 0.4, 0.01 * 0.2 \rangle \approx \langle 0.77, 0.22, 0.01 \rangle$. Hence $P(dog \mid loudnoise) = 0.77$. Which BTW is $> P(dog \mid likeschappi) = 0.64$.

Example



Naive Enumeration:

$$P(A, D = f) =$$

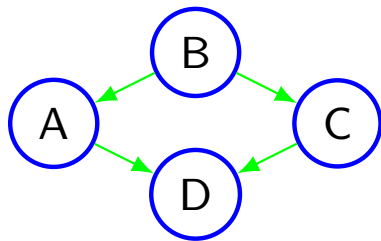
$$\sum_{b \in \{t, f\}} \sum_{c \in \{t, f\}} P(B = b, A, C = c, D = f) =$$

$$\sum_{b \in \{t, f\}} \sum_{c \in \{t, f\}} P(B = b)P(A \mid B = b)P(C = c \mid B = b)P(D = f \mid A, C = c)$$

Observation 1: $P(B = b)P(A \mid B = b)$ does not depend on C , so

$$= \sum_{b \in \{t, f\}} P(B = b)P(A \mid B = b) \sum_{c \in \{t, f\}} P(C = c \mid B = b)P(D = f \mid A, C = c)$$

Example continued



B		
	t	f
	.5	.5

B	A	
	t	f
t	.7	.3
f	.1	.9

B	C	
	t	f
t	.7	.3
f	.2	.8

A	C	D	
		t	f
t	t	.9	.1
t	f	.7	.3
f	t	.8	.2
f	f	.4	.6

Observation 2: Let's precompute first factor F_1

$$\sum_b P(B = b) P(A \mid B = b) \underbrace{\sum_c P(C = c \mid B = b) P(D = f \mid A, C = c)}_{F_1} =$$

$$\sum_b P(B = b) P(A \mid B = b) F_1(B = b, A) = F_2(A)$$

Is F_1 a single numerical value? No!, a table $F_1(A, B)$ because it takes different values for each value of variables A and B .

where

B	C	
	t	f
t	.7	.3
f	.2	.8

A	C	D	
		t	f
t	t	.9	.1
t	f	.7	.3
f	t	.8	.2
f	f	.4	.6

 $\mapsto$

b	a	$F_1(B, A)$
t	t	$.7 \cdot .1 + .3 \cdot .3 = .16$
t	f	$.7 \cdot .2 + .3 \cdot .6 = .32$
f	t	$.2 \cdot .1 + .8 \cdot .3 = .26$
f	f	$.2 \cdot .2 + .8 \cdot .6 = .52$

and

	B	
	t	f
	.5	.5

B	A	
	t	f
t	.7	.3
f	.1	.9

b	a	$F_1(B, A)$
t	t	.16
⋮	⋮	⋮

 $\mapsto$

a	$F_2(A)$
t	...
f	...

Variable Elimination

The procedure operates on **factors**: functions of subsets of variables

Definition (Factor). A **factor** $F(V_1, \dots, V_k)$ is a function mapping each combination of values of the variables $V_1, \dots, V_k$ to a number.

→ Factors are not necessarily CPTs as numbers do not need to sum up to 1.

Required operations on factors:

- **Restriction** (setting selected variables to specific values)
- **Multiplication** (join tables and multiply the probabilities)
- **Marginalization** (summing out selected variables)

Restriction (of variable D to value $D = f$):

		$F(A, C, D)$	
		D	
A	C	t	f
t	t	.9	.1
t	f	.7	.3
f	t	.8	.2
f	f	.4	.6

→

A	C	$F_2(A, C)$
t	t	.1
t	f	.3
f	t	.2
f	f	.6

Variable Elimination

The procedure operates on **factors**: functions of subsets of variables

Definition (Factor). A **factor** $F(V_1, \dots, V_k)$ is a function mapping each combination of values of the variables $V_1, \dots, V_k$ to a number.

→ Factors are not necessarily CPTs as numbers do not need to sum up to 1.

Required operations on factors:

- **Restriction** (setting selected variables to specific values)
- **Multiplication** (join tables and multiply the probabilities)
- **Marginalization** (summing out selected variables)

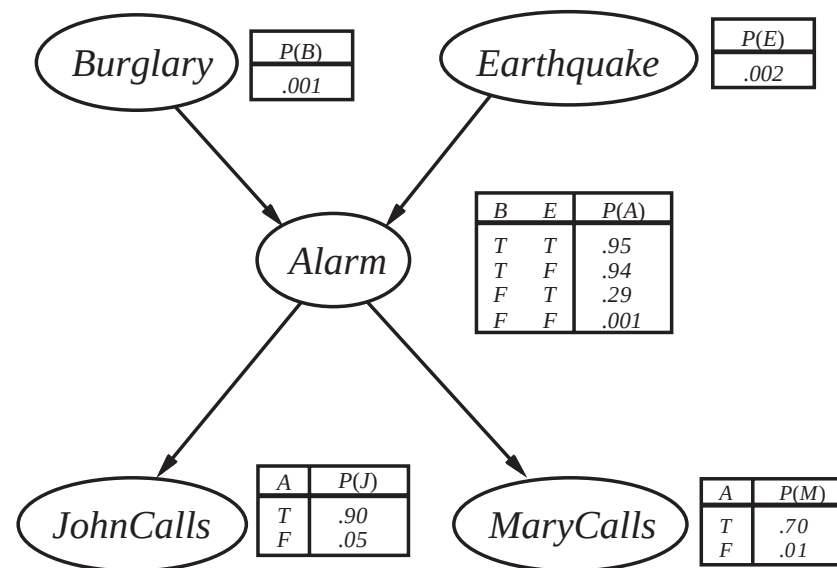
Multiplication + Marginalization (of variable C):

B	C	$F_1(B, C)$	A	C	$F_2(A, C)$		b	a	$F(B, A)$
t	t	.7	t	t	.1	→	t	t	$.7 \cdot .1 + .3 \cdot .3 = .16$
t	f	.3	t	f	.3		t	f	$.7 \cdot .2 + .3 \cdot .6 = .32$
f	t	.2	f	t	.2		f	t	$.2 \cdot .1 + .8 \cdot .3 = .26$
f	f	.8	f	f	.6		f	f	$.2 \cdot .2 + .8 \cdot .6 = .52$

Variable Elimination Algorithm

1. Factors = CPT in BN
2. For all variables in the evidence, restrict all factors with the observed value
3. Fix any order of the remaining variables, $X_1, \dots, X_n$.
4. **for** $i := 1, \dots, n$ **do**
 - a. $\mathcal{F} = \{F \mid F \in \text{Factors}, F \text{ depends on } X_i\}$
 - b. $T = \pi_{F \in \mathcal{F}}$ (Compute product of factors)
 - c. $F_{new} = \text{Marginalize}(T, X_i)$
 - d. $\text{Factors} = \text{Factors} \setminus \mathcal{F} \cup \{N\}$ (Replace all factors $\mathcal{F}$ by new factor N)

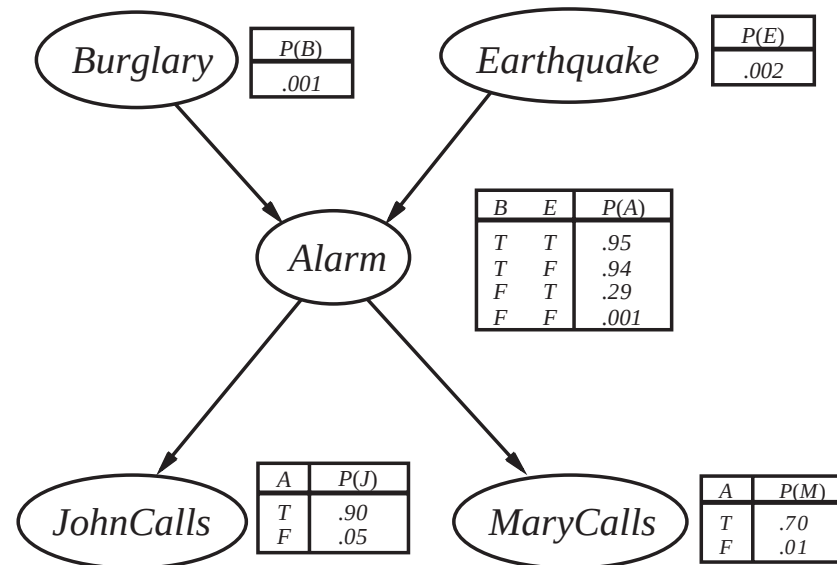
Variable Elimination Algorithm: Alarm Example



Bad ordering for computing $P(M, B = t)$: **A**, **J**, **E** See lecture slides for animated example

1. Factors = CPT in BN
2. For all variables in the evidence, restrict all factors with the observed value
3. Fix any order of the remaining variables, $X_1, \dots, X_n$.
4. **for** $i := 1, \dots, n$ **do**
 - a. $\mathcal{F} = \{F \mid F \in \text{Factors}, F \text{ depends on } X_i\}$
 - b. $T = \pi_{F \in \mathcal{F}}$ (Compute product of factors)
 - c. $F_{\text{new}} = \text{Marginalize}(T, X_i)$
 - d. Factors = Factors $\setminus \mathcal{F} \cup \{N\}$ (Replace all factors $\mathcal{F}$ by new factor N)

Alarm Example

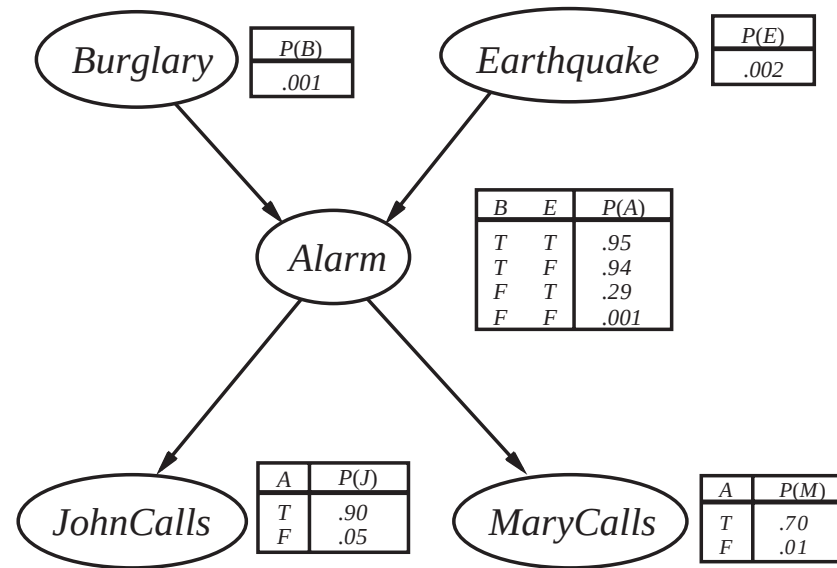


Bad ordering for computing $P(MC, B = t)$: A, J, E

$$\begin{aligned}
 & \sum_{eq \in \{t, f\}} \sum_{jc \in \{t, f\}} \sum_{a \in \{t, f\}} P(B = t) P(EQ = eq) P(A = a \mid B = t, EQ = eq) P(JC = jc \mid A = a) P(MC \mid A = a) = \\
 & \sum_{eq \in \{t, f\}} \sum_{jc \in \{t, f\}} P(B = t) P(EQ = eq) F_1(eq, jc, MC) = \\
 & \sum_{eq \in \{t, f\}} P(B = t) P(EQ = eq) F_2(eq, MC) = \\
 & P(B = t) F_3(MC)
 \end{aligned}$$

Largest factor (F_1) is function of 3 variables!

Alarm Example continued



Question!

What's the maximum number of columns of the largest table generated by variable elimination under ordering: J,E,A?

(A): 1

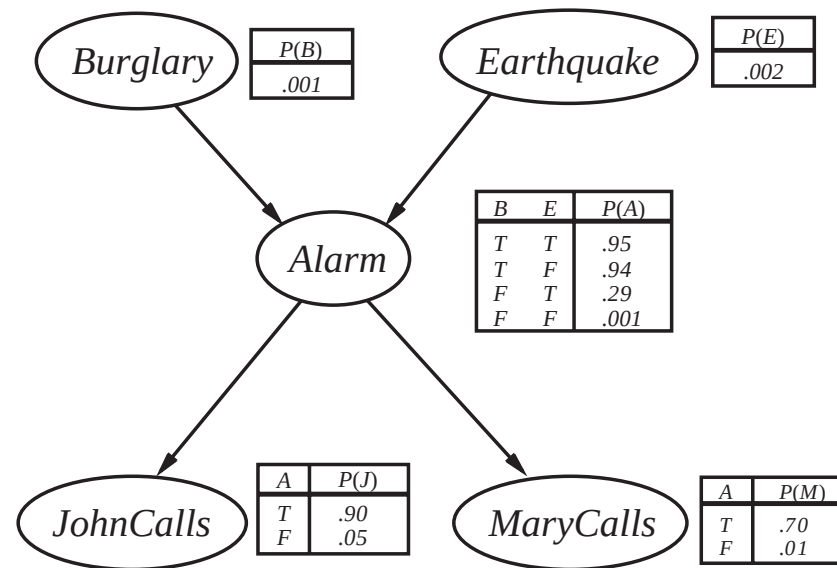
(B): 2

(C): 3

(D): 4

Largest factor $F_2(A, E)$ is function of 2 variables!

Alarm Example continued



Good ordering for computing $P(MC, B = t)$:

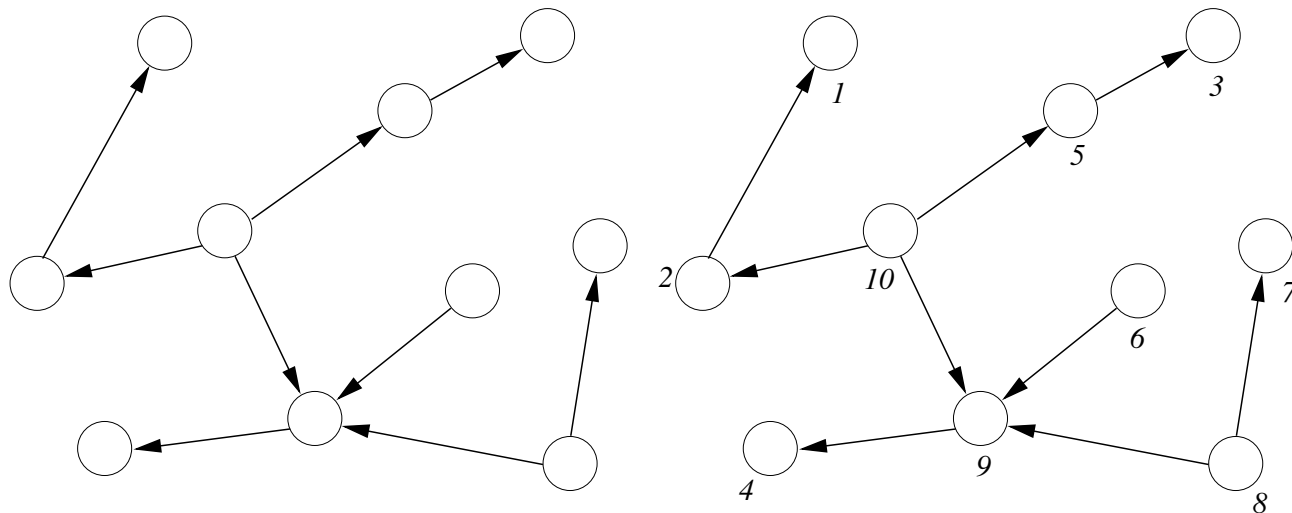
$$\begin{aligned}
 & \sum_{a \in \{t, f\}} \sum_{eq \in \{t, f\}} \sum_{jc \in \{t, f\}} P(B = t) P(EQ = eq) P(A = a \mid B = t, EQ = eq) P(JC = jc \mid A = a) P(MC \mid A = a) = \\
 & \sum_{a \in \{t, f\}} \sum_{eq \in \{t, f\}} P(B = t) P(EQ = eq) P(A = a \mid B = t, EQ = eq) P(MC \mid A = a) F_1(a) = \\
 & \sum_{a \in \{t, f\}} P(B = t) P(MC \mid A = a) F_1(a) F_2(a) = \\
 & P(B = t) F_3(MC)
 \end{aligned}$$

Largest factor ($P(A \mid B = t, EQ)$) is function of 2 variables!

Variable Elimination Runtime

An important easy case:

- A graph is called **singly connected**, or a **polytree**, if there is at most one undirected path between any two nodes in the graph.



→ Is our BN for Mary & John a polytree? Yes.

For singly connected network: any elimination order that “peels” variables from outside will only create factors of only one variable.

The complexity of inference is therefore **linear in the total size of the network** (= combined size of all conditional probability tables).